Markov Chain Lock Watermarking: A Calibrated Policy-to-Configuration Map for EU AI Act Article 50

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Abstract

EU AI Act Article 50 requires machine-readable marking of AI-generated content but leaves two decisive parameters – false positive rate and text-quality cost – to be chosen opaquely by deployers rather than specified by regulators. We present *Markov Chain Lock* (MCL) watermarking, an active watermarking scheme that partitions the vocabulary into S states via SHA-256 hashing and constrains generation to follow a clockwork transition $s \to (s+1) \bmod S$ at selected (low-entropy) positions. Detection is model-free, runs in $O(n)$ hash operations, and admits Hoeffding-style exponential false positive bounds – making third-party audit feasible without LLM access. The central result is a calibration formula $S^\star = \lceil 4z_\alpha^2/(\rho^2(n-1)) + 1\rceil$ that maps a target FPR α, text length n, and watermark fraction ρ to a minimum state count, turning Article 50 compliance from an opaque deployer choice into an auditable policy-to-configuration map. We additionally prove a universal robustness threshold $\delta^\star = 1 - 1/\sqrt{2} \approx 29.3\%$ independent of (S, ρ, n), derive a Neyman–Pearson optimal entropy-weighted detector, and formalize a $1/S$ self-healing surplus under adversarial modification. A critical experiment tests whether low-entropy tokens survive paraphrasing at higher rates than high-entropy ones – inverting the SWEET/EWD heuristic – and decides whether the entropy gate is purely a quality improvement or also a robustness one.

1. Introduction

Regulatory regimes are now committing to machine-readable marking of AI-generated content. EU AI Act Article 50 (European Parliament and Council, 2024), with enforcement beginning August 2026, mandates that providers of generative AI systems embed detectable signals in their outputs; parallel requirements appear in the OECD Hiroshima Process (OECD, 2024) and in the EU's draft Code of Practice on Transparency of AI-Generated Content (European Commission AI Office, 2025). The question these regimes leave unspecified is *what counts as compliance*: at what false positive rate, for what text length, at what quality cost.

The tradeoff that no one is naming. Active watermarking schemes sit on a multidimensional frontier among detection power, robustness to modification, text-quality cost, and generation-time complexity. Current deployments choose a point on this frontier privately. A regulator who writes "outputs shall be marked" has no mechanism to specify *where* on the frontier to sit, and an auditor has no basis on which to evaluate a deployer's choice. The consequence is that Article 50 is straightforward to comply with *nominally* (mark something, anything) and essentially unauditable *substantively*.

Existing watermarking approaches do not fix this. Passive detection (Mitchell et al., 2023; Tian, 2023) reads statistical traces in the text and suffers high false positive rates on technical prose, with no controllable frontier at all. Green–red-list watermarking (Kirchenbauer et al., 2023) biases logits during generation but requires temperature sampling and degrades under paraphrase (Krishna et al., 2023). Schemes establishing the undetectability frontier (Christ et al., 2024) are informative bounds but not directly deployable. None expose a calibration map from regulatory parameters to deployed configuration.

Markov Chain Lock watermarking. We present *Markov Chain Lock* (MCL) watermarking: a scheme that partitions the vocabulary into S states via $\sigma_k(t) = H(k \,\|\, t) \bmod S$ (SHA-256) and constrains generation to follow a clockwork transition $s \to (s+1) \bmod S$. Watermarking is *entropy-gated*: the constraint applies only at positions where the model is confident, leaving high-entropy positions unconstrained so that the generator's expressive budget is spent on positions where fluency is decided. The scheme has three properties that suit it to operationalising Article 50:

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- **Model-free detection.** Verification maps each token to a state via the shared key and counts valid transitions in $O(n)$ hash operations. A third-party auditor needs neither the original model nor a proxy LM.
- **Exponential FPR.** A watermarked sequence has score 1 by construction; random sequences have score $1/S$ with Hoeffding bound $\exp(-2(n-1)(\tau-1/S)^2)$ on the FPR.
- **Greedy decoding compatibility.** The constraint is a logit mask, not a temperature bias, so generation remains deterministic.

Calibration: the regulatory knob. The core analytical result is a closed-form mapping from deployment regime to configuration. Under the null, the detection z-score is $z = (\rho/2)\sqrt{(S-1)(n-1)}$ where ρ is the watermark fraction. Inverting for the minimum S at which a target FPR α is met gives

$$S^\star(n,\rho,\alpha) = \left\lceil \frac{4z_\alpha^2}{\rho^2(n-1)} + 1 \right\rceil. \qquad (1)$$

A regulator choosing a target FPR α and a text-length floor $n_{\min}$ reads off a unique minimum configuration; a deployer can only adjust ρ upward, trading quality for margin, but cannot undercut detection. This is, we argue, the regulatory knob Article 50 has been missing.

Contributions.

1. An end-to-end active watermarking scheme (MCL) with SHA-256 state partitioning, clockwork transitions, entropy-gated embedding (§4), and model-free $O(n)$ detection.
2. A calibration formula (1) derived from a generalized detection bound $z = (\rho/2)\sqrt{(S-1)(n-1)}$ (Thm. 5.1), mapping regulatory parameters to deployed configurations.
3. A universal robustness threshold $\delta^\star = 1 - 1/\sqrt{2} \approx 29.3\%$ (Thm. 5.3) at which detection crosses the midpoint, independent of S, ρ, and n – a scheme-independent floor regulators can reference directly.
4. A Neyman–Pearson optimal entropy-weighted detector (Thm. 5.5) that strictly dominates the model-free detector when the entropy profile is observable, and a self-healing theorem (Thm. 5.6) showing each adversarial modification contributes $1/S$ in expectation to the valid-transition count.
5. An experimental program (§6) with one critical test: whether low-entropy tokens survive paraphrasing at higher rates than high-entropy ones. If yes, entropy gating is also a robustness win and the SWEET (Lee et al., 2024) / EWD (Lu et al., 2024) heuristic (watermark the high-entropy tokens) is inverted; if no, we revise the robustness claim accordingly.

Governance framing. The contribution is as much about *operationalising* Article 50 as it is about watermarking. The calibration formula turns an opaque deployer choice into a two-number policy specification $(\alpha, n_{\min})$ that an auditor can check. The model-free $O(n)$ detector makes that audit feasible without model access. And the universal $\delta^\star$ gives regulators a scheme-independent robustness benchmark. §7 lays out a minimum viable Article 50 specification in concrete numbers.

2. Related Work

Passive detection. DetectGPT (Mitchell et al., 2023) exploits log-probability curvature; GPTZero (Tian, 2023) combines perplexity and burstiness. Both read statistical traces in existing text, with no controllable false-positive frontier and sharp degradation under paraphrase (Krishna et al., 2023; Sadasivan et al., 2023). They cannot serve as the technical substrate for Article 50 because neither regulator nor deployer can set the detection regime in advance.

Active watermarking. Green–red-list watermarking (Kirchenbauer et al., 2023) biases next-token logits toward a hashed green list and detects via a z-test on green-token frequency. The scheme is well-understood but requires temperature sampling, degrades under paraphrase, and keeps the detection-quality tradeoff inside a single bias magnitude with no analytic inversion back to a regulatory parameter. Christ et al. (2024) establish the formal quality–security frontier for undetectable watermarks; the result bounds what any scheme can achieve but does not yield a deployable construction. Aaronson (2023) and the SemStamp line embed signals in semantic-level features and require LLM queries at detection time, foreclosing lightweight third-party audit.

Entropy-adaptive schemes. SWEET (Lee et al., 2024) and EWD (Lu et al., 2024) restrict watermarking to *high-entropy* positions, arguing the scheme has the most room to inject signal where the model is least committed. Our scheme inverts this policy: MCL watermarks at *low*-entropy positions, trading on the intuition that near-deterministic tokens survive meaning-preserving rewrites at higher rates. We formalize and test this inversion (§6).

Technical AI governance. Article 50 of the EU AI Act (European Parliament and Council, 2024) and the EU's draft Code of Practice on Transparency (European Commission AI Office, 2025) require machine-readable marking of AI-generated content; the OECD Hiroshima Process (OECD, 2024) provides a parallel international framework. These instruments leave the detection regime (FPR, quality floor, robustness) implicit. MCL's contribution, beyond the watermarking scheme itself, is that it surfaces these parameters analytically, producing a policy-to-configuration map auditable by third parties (§7).

3. Preliminaries

Let $\mathcal{V}$ denote the vocabulary of an autoregressive LLM $\mathcal{M}$, $V = |\mathcal{V}|$. At step i the LM defines a next-token distribution p_i conditional on the prefix $t_1 \dots t_{i-1}$; its entropy is $H_i = -\sum_t p_i(t) \log p_i(t)$. We assume throughout that the verifier and the deployer share a secret key $k \in \{0,1\}^*$ and use the same tokenizer. Let $H : \{0,1\}^* \to \{0,1\}^{256}$ be SHA-256, modelled as a random oracle in the security analysis (5.4). We write $[n] = \{1, \dots, n\}$ and z_α for the one-sided standard-normal quantile at level α.

4. The MCL Scheme

MCL has three components: a cryptographic partition of the vocabulary into states, a clockwork transition over states, and an entropy gate selecting where to enforce the transition.

4.1. State partition

Definition 4.1 (State assignment)**.** For key k and state count $S \in \mathbb{N}$, define $\sigma_k(t) = H(k \,\|\, t) \bmod S$. This induces a partition $\mathcal{V} = \bigsqcup_{s=0}^{S-1} \mathcal{V}_s$ with $\mathcal{V}_s = \{t \in \mathcal{V} : \sigma_k(t) = s\}$.

Under the random oracle, $\mathbb{E}[|\mathcal{V}_s|] = V/S$ and $\mathrm{Var}(|\mathcal{V}_s|) = V \cdot \frac{S-1}{S^2}$, so the partition is approximately balanced for realistic V.

4.2. Clockwork transitions

Definition 4.2 (Clockwork matrix)**.** The S-state clockwork transition matrix is $\mathbf{T}^{\mathrm{clk}}_{ij} = \mathbb{1}[j \equiv i + 1 \pmod S]$.

The clockwork is deterministic (each state has exactly one legal successor), periodic of period S, and has uniform stationary distribution $\pi^\star = (1/S, \dots, 1/S)$. Among S-state irreducible deterministic transition matrices, clockwork minimises the random-baseline validity $1/S$ – maximising the detection gap.

4.3. Entropy gate

At each generation step, MCL decides whether to enforce the clockwork transition based on the model's confidence at that position.

Definition 4.3 (Entropy gate)**.** Given an entropy threshold τ_H, the gate at position i is $g_i = \mathbb{1}[H_i < \tau_H]$. The watermark fraction is the expected proportion of gated positions, $\rho = \mathbb{E}_i[g_i] \in [0, 1]$.

Setting $\tau_H = \infty$ recovers the non-gated variant ($\rho = 1$). Lower τ_H concentrates the watermark at near-deterministic positions where the LM's own distribution is narrow and the clockwork mask is cheap.

Algorithm 1 MCL Watermark Embedding

Input: prompt p, key k, states S, entropy threshold τ_H, max tokens N
$\mathbf{t} \leftarrow \mathrm{Tokenize}(p)$; $s \leftarrow \sigma_k(\mathbf{t}_{-1})$
for $i = 1$ **to** N **do**
  $p_i \leftarrow \mathrm{softmax}(\mathcal{M}(\mathbf{t}))$; $H_i \leftarrow -\sum_t p_i(t) \log p_i(t)$
  if $H_i < \tau_H$ **then**
    $s' \leftarrow (s + 1) \bmod S$
    $t_i \leftarrow \arg\max_{t \in \mathcal{V}_{s'}} p_i(t)$
  else
    $t_i \leftarrow \arg\max_t p_i(t)$
  end if
  $\mathbf{t} \leftarrow \mathbf{t} \,\|\, t_i$; $s \leftarrow \sigma_k(t_i)$
  if $t_i = \mathtt{EOS}$ **then**
    break
  end if
end for
return $\mathrm{Decode}(\mathbf{t})$

Algorithm 2 MCL Watermark Detection (model-free)

Input: text x, key k, states S, threshold τ
$\mathbf{t} \leftarrow \mathrm{Tokenize}(x)$; $c \leftarrow 0$
for $i = 1$ **to** $n - 1$ **do**
  if $\sigma_k(t_{i+1}) \equiv \sigma_k(t_i) + 1 \pmod S$ **then**
    $c \leftarrow c + 1$
  end if
end for
$\phi \leftarrow c/(n - 1)$; $z \leftarrow (\phi - 1/S)/\sqrt{(1/S)(1 - 1/S)/(n - 1)}$
return $(\phi > \tau,\ \phi,\ 1 - \Phi(z))$

4.4. Embedding

1 requires no sampling and no temperature parameter. Each step does one LM forward pass and one $O(V)$ logit mask; total cost $O(N(V + T_{\mathrm{LM}}))$ for N generated tokens.

4.5. Detection

Definition 4.4 (Fingerprint score)**.** For a token sequence $\mathbf{t} = (t_1, \dots, t_n)$, the *fingerprint score* is the fraction of valid clockwork transitions:

$$\phi(\mathbf{t}) = \frac{1}{n-1} \sum_{i=1}^{n-1} \mathbf{T}^{\mathrm{clk}}_{\sigma_k(t_i), \sigma_k(t_{i+1})}. \tag{2}$$

2 runs in $O(n)$ hash operations with no LM access. This is the property that makes MCL a candidate substrate for third-party audit.

5. Theoretical Guarantees

We state five theorems about MCL; full proofs are in A.

Theorem 5.1 (Detection bound and calibration). *Let $\mathbf{t}^w$ be an MCL-watermarked sequence of length n with watermark fraction ρ, and $\mathbf{t}^r$ an i.i.d. random sequence over $\mathcal{V}$. For the clockwork matrix with S states and threshold $\tau \in (1/S, 1)$:*

$$\mathbb{E}[\phi(\mathbf{t}^w)] = \tfrac{1}{S} + \rho\tfrac{S-1}{S}, \tag{3}$$

$$\mathbb{E}[\phi(\mathbf{t}^r)] = \tfrac{1}{S}, \tag{4}$$

$$\mathrm{FPR} \le \exp\bigl(-2(n-1)(\tau - 1/S)^2\bigr), \tag{5}$$

$$z_{\mathrm{FPR}}(\rho, S, n) = \tfrac{\rho}{2}\sqrt{(S-1)(n-1)}. \tag{6}$$

Inverting for the minimum S achieving detection at FPR α:

$$S^\star(n, \rho, \alpha) = \left\lceil \frac{4z_\alpha^2}{\rho^2(n-1)} + 1 \right\rceil. \tag{7}$$

7 is the calibration formula. It maps a target FPR α, a text-length floor n, and a watermark fraction ρ to a minimum state count $S^\star$. At $\alpha = 10^{-6}$, $\rho = 0.5$: $S^\star = 6$ for $n = 100$, $S^\star = 3$ for $n = 200$, $S^\star = 2$ for $n \ge 380$. For long texts, two states suffice.

Theorem 5.2 (Quality bound). *At a gated position, the KL divergence between the MCL-constrained distribution and the unconstrained LM distribution satisfies $D_{\mathrm{KL}}(P_{\mathrm{MCL}} \,\|\, p) = \log(1/Z_s)$, where $Z_s = p(\mathcal{V}_{(s+1) \bmod S})$ is the probability mass the LM assigned to the target partition. Under hash uniformity, $\mathbb{E}[Z_s] \ge 1/S$, giving the position-wise bound $\mathbb{E}[D_{\mathrm{KL}}] \le \log S$ and the per-token average $\mathbb{E}[D_{\mathrm{KL}}]_{tok} \le \rho \log S$.*

Theorem 5.3 (Universal robustness threshold). *Under uniform random modification of a fraction δ of tokens,*

$$\mathbb{E}[\phi \mid \mathrm{attack}] = \tfrac{1}{S} + \rho(1-\delta)^2\tfrac{S-1}{S}. \tag{8}$$

At the midpoint threshold $\tau = \frac{1}{2}\bigl(1/S + \mathbb{E}[\phi \mid \mathrm{WM}]\bigr)$, the critical modification fraction is

$$\delta^\star = 1 - 1/\sqrt{2} \approx 0.293, \tag{9}$$

independent of S, ρ, and n.

Theorem 5.4 (Information-theoretic security). *Under the random oracle model for H, any polynomial-time distinguisher $\mathcal{A}$ without key k satisfies*

$$\mathrm{Adv}(\mathcal{A}) = \bigl|\Pr[\mathcal{A}(\mathbf{t}^w) = 1] - \Pr[\mathcal{A}(\mathbf{t}^r) = 1]\bigr| \le \mathrm{negl}(256). \tag{10}$$

Moreover, $\mathcal{A}$ cannot recover the gate pattern (g_i) from tokens alone.

Theorem 5.5 (Entropy-weighted detector is Neyman—Pearson optimal). *Suppose the verifier has access to the LM's next-token distribution at each position. Let $X_i = \mathbb{1}\bigl[\sigma_k(t_{i+1}) \equiv \sigma_k(t_i) + 1 \pmod S\bigr]$ and $\pi_i = P(X_i = 1 \mid$ watermarked$)$. The log-likelihood ratio test*

$$\Lambda = \sum_{i=1}^{n-1} \Bigl[X_i \log \tfrac{\pi_i S}{1} + (1 - X_i) \log \tfrac{1-\pi_i}{1-1/S}\Bigr] \tag{11}$$

is Neyman–Pearson optimal among all size-α tests for distinguishing watermarked from random sequences under the known entropy profile.

Theorem 5.6 (Self-healing). *Against any adversary who modifies tokens without oracle access to σ_k, each modified token contributes $1/S$ in expectation to the valid-transition count. Equivalently, the self-healing surplus over the null is*

$$\Sigma(\delta) = \tfrac{1}{S}\delta(2 - \delta). \tag{12}$$

The surplus is strategy-independent*: the attacker cannot concentrate modifications to reduce it below $1/S$ per modified token.*

Interpretation. 5.1 gives the central regulatory knob: detection scales as $\sqrt{n-1}$, so length trades directly against states or against watermark fraction. 5.3 gives a scheme-independent robustness floor – useful as a reference number in Article 50 guidance. 5.5 tells the verifier with model access exactly how much detection power they stand to gain. 5.6 is the mathematical backbone of robustness: modifications cannot drive detection below the null baseline.

6. Experimental Validation

We validate MCL on a curated benchmark of 173 Wikipedia concepts covering notable figures, historical events, scientific phenomena, geographical entities, technology, and popular culture. Each concept yields a prompt of the form *"Explain [concept] in a comprehensive way."* Generation uses greedy decoding for watermarked outputs and nucleus sampling (T=0.7, p=0.9) for the unwatermarked baseline. Primary model: Llama-3.2-3B-Instruct; secondary models for cross-model runs: Mistral-7B, GPT2-XL. Secret key `curated_wiki_dataset_2024`, seed 42, max 200 tokens.

We report six experiments. E1–E3 validate the calibration formula and the quality/robustness theorems directly. E4 is the critical test of the low-entropy-robustness hypothesis. E5–E6 validate self-healing and HDD.

E1: Calibration. We verify that $S^\star(n, \rho, \alpha)$ predicts empirical TPR and FPR. Configurations span $S \in \{2, 3, 4, 5, 7, 9\}$, entropy thresholds $\tau_H \in \{1.5, 2.0, 2.5, 3.0, 3.5\}$, text lengths $n \in \{200, 500\}$. The prediction is that $S \ge S^\star(n, \rho, \alpha)$ achieves $\mathrm{TPR} \approx 1$ and $\mathrm{FPR} \approx \alpha$; $S < S^\star$ misses the target.

E2: Quality. Matched-detection comparison between entropy-gated configurations on the calibration curve and a non-gated reference ($\rho = 1$, $S = 7$). At equal detection power (equal z), does gated MCL have lower perplexity than the non-gated reference? 5.2 predicts yes for sufficiently long texts.

E3: Robustness. Seven attacks: (i) random word replacement, $\delta \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$; (ii) WordNet synonym substitution; (iii) DIPPER paraphrase; (iv) back-translation (en→fr→en, en→zh→en); (v) homoglyph perturbation; (vi) word insertion; (vii) word deletion. Primary metric: TPR at $p < 0.01$ after attack. 5.3 predicts $\delta^\star \approx 0.293$ universal for (i); (iii)–(iv) are the real-world rewrite tests that load on E4.

E4: Entropy–survival (critical). For 500 watermarked texts, record the per-position entropy H_i during generation. Paraphrase each text with DIPPER, GPT-rewrite, and back-translation. Align original and rewritten tokens via edit distance. For each entropy bin, compute the empirical *token survival rate* $\Pr[t'_i = t_i \mid H_i]$. The hypothesis is that survival rate decreases monotonically with H_i and exceeds 0.6 for $H_i < 1.0$. **Falsification would invalidate the low-entropy-robustness claim** and force a revision of §7.

E5: Self-healing. For $S \in \{2, 3, 5, 7\}$ and $\delta \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$, compare the observed ϕ after random replacement to the closed form in 5.3, and verify $\delta^\star \approx 0.293$ is universal across S.

E6: HDD vs. model-free detection. ROC comparison of three detectors: (a) oracle-entropy HDD, (b) proxy-entropy HDD using GPT-2 as a cheap verifier for Llama-3.2, and (c) model-free detection. 5.5 predicts (a) > (c); the open empirical question is how much (b) concedes relative to (a).

Reference point. A non-gated configuration at S=7, ρ=1 reaches 100% TPR, 0% FPR, perplexity 4.20; unwatermarked baseline perplexity is ~ 3.0. 1 in B gives the full sweep.

7. Discussion

7.1. Policy operationalisation

We read Article 50 as a two-parameter specification problem: *what false positive rate must marked content guarantee, and at what minimum text length?* Let $(\alpha, n_{\min})$ be this specification. A deployer faces three choices:

1. Watermark fraction ρ (controls quality cost via 5.2).
2. State count S (controls detection power via 5.1).
3. Entropy threshold τ_H (operationally sets ρ; higher τ_H $\Rightarrow$ higher $\rho \Rightarrow$ more quality cost).

The calibration formula (7) reduces this to a single constraint on (S, ρ): any (S, ρ) pair with $S \geq S^\star(n_{\min}, \rho, \alpha)$ satisfies the specification. Auditing a deployment then requires measuring three quantities from samples – the effective ρ, the key-derived S, and the text-length distribution – and checking them against the specification. No LM access needed.

A minimum viable Article 50 rule. As a concrete illustration: set $\alpha = 10^{-6}$ and $n_{\min} = 200$. Then (7) with $\rho = 0.5$ gives $S^\star = 3$. A regulator writing "outputs shall be markable at false-positive rate 10^{-6} on passages of 200 tokens or more" would, under MCL, translate to "use at least 3 states at watermark fraction 0.5". This is a concrete, auditable, minimum-viable rule. Stronger rules (higher α, lower $n_{\min}$) push $S^\star$ up monotonically.

7.2. When to use what

For short texts ($n < 100$), detection requires large S and the calibration sits at or above the non-gated baseline. For long texts ($n > 300$), gated MCL at $S \in \{2, 3\}$ with $\rho \approx 0.5$ dominates the non-gated reference: equal detection, lower perplexity (5.2). The governance-relevant regime – long-form generated content subject to Article 50 – is the one where gating pays off.

7.3. Limitations

Tokenizer alignment. Detection assumes the verifier uses the same tokenizer as the generator. Adversaries who retokenize via a different vocabulary can break detection without altering surface text. This is a known limitation of token-level watermarking schemes.

Key management. If k is leaked, all past and future watermarks are exposed and replayable. MCL as presented provides no forward security; key rotation schemes are straightforward in principle but not evaluated here.

The fixed-point hypothesis. The robustness argument for gating rests on E4. 5.3 gives an unconditional $\delta^\star$ under *uniform random* modification. Real rewriters are not uniform: they concentrate changes where the LM allows. The low-entropy-robustness claim is that rewriters preserve low-entropy tokens at rate $\geq q_i$ (the LM top-probability). We test this directly; a negative result would preserve 5.3 but rescind the robustness motivation for gating specifically.

Quality bound is crude. $\mathbb{E}[D_{\mathrm{KL}}] \leq \rho \log S$ is an upper bound, not a tight characterisation. The actual perplexity cost depends on the mass the LM concentrates in the target partition at each gated position, which is structured (not uniform) and model-specific.

Greedy decoding assumed. MCL's determinism is compatible with greedy decoding but not with temperature sampling, which a deployer may require for other reasons. Temperature sampling weakens the quality bound and does not affect detection.

8. Conclusion

MCL watermarking makes the Article 50 detection-quality tradeoff analytically explicit via the calibration formula $S^\star = \lceil 4z_\alpha^2/(\rho^2(n-1)) + 1\rceil$, and pins down its robustness floor at a scheme-independent $\delta^\star \approx 29.3\%$. Detection is model-free and runs in $O(n)$ hash operations, enabling third-party audit without LM access. The remaining empirical question is whether low-entropy tokens enjoy higher survival rates under meaning-preserving rewrite than high-entropy ones – inverting the conventional entropy-adaptive heuristic. The answer, produced by E4, decides whether entropy gating is a pure quality win or also a robustness win. Either way, the policy-to-configuration map does not depend on the outcome: MCL remains the first watermarking scheme to expose Article 50's implicit parameters as a calibration law.

Impact Statement

This paper develops watermarking infrastructure for LLM-generated content in service of AI governance regimes (EU AI Act Article 50, OECD Hiroshima Process). Watermarking has dual-use properties: it supports provenance and accountability, but a deployed watermark can also be used to track authorship of content that users believed was untraceable, and stronger detection invites stronger attacks. Our model-free detector reduces verification-side compute and enables third-party audit without LM access, which we view as governance-positive. Any deployment should disclose its detection regime (target FPR, text-length floor, watermark fraction) to the public so that users understand under what conditions their outputs are, and are not, marked.

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A. Proofs

A.1. Proof of 5.1

Let $Z_i = \mathbf{T}^{\text{clk}}_{\sigma_k(t_i),\sigma_k(t_{i+1})}$ be the validity indicator at step i.

Null mean. Under the random oracle, σ_k is uniform over $\mathcal{S}$ independently across distinct tokens, so $Z_i \sim \text{Bernoulli}(1/S)$ for random input. Hence $\mathbb{E}[\phi \mid \text{null}] = 1/S$.

Watermark mean. At a gated position, 1 sets $\sigma_k(t_{i+1}) = (\sigma_k(t_i) + 1) \bmod S$, so $Z_i = 1$. At an ungated position, $Z_i \sim \text{Bernoulli}(1/S)$. Hence $\mathbb{E}[\phi \mid \text{WM}] = \rho \cdot 1 + (1-\rho) \cdot 1/S = 1/S + \rho(S-1)/S$.

FPR bound. By Hoeffding's inequality for bounded i.i.d. random variables, $\Pr(\phi \geq \tau) \leq \exp(-2(n-1)(\tau - 1/S)^2)$ when $\tau > 1/S$.

z-score. The null has variance $(1/S)(1-1/S)/(n-1)$. The signed mean gap between watermarked and null is $\rho(S-1)/S$. Dividing:

$$z = \frac{\rho(S-1)/S}{\sqrt{(1/S)(1-1/S)/(n-1)}} = \rho\sqrt{\frac{(S-1)^2}{S^2} \cdot \frac{S^2}{S-1}(n-1)} = \rho\sqrt{(S-1)(n-1)}.$$

A factor of 2 discrepancy with 7 arises from using the midpoint threshold in the calibration; substituting $\tau = 1/S + \frac{\rho}{2}(S-1)/S$ recovers $z_{\text{FPR}} = (\rho/2)\sqrt{(S-1)(n-1)}$.

Calibration. Solving $z_{\text{FPR}} \geq z_\alpha$ for S: $(\rho/2)\sqrt{(S-1)(n-1)} \geq z_\alpha$, i.e. $S \geq 4z_\alpha^2/(\rho^2(n-1)) + 1$. Taking ceiling gives (7). □

A.2. Proof of 5.2

At a gated position with current state s, the MCL-constrained distribution is $P_{\text{MCL}}(t) = p(t)/Z_s$ for $t \in \mathcal{V}_{(s+1) \bmod S}$ and 0 otherwise, where $Z_s = p(\mathcal{V}_{(s+1) \bmod S})$. By direct computation,

$$D_{\text{KL}}(P_{\text{MCL}} \,\|\, p) = \sum_{t \in \mathcal{V}_{(s+1) \bmod S}} \frac{p(t)}{Z_s} \log \frac{1}{Z_s} = \log(1/Z_s).$$

Under hash uniformity, $\mathbb{E}[Z_s] \geq 1/S$ in expectation over the key, giving $\mathbb{E}[D_{\text{KL}}] \leq \log S$ at each gated position. Averaging over positions with gate rate ρ: $\mathbb{E}[D_{\text{KL}}]_{\text{tok}} \leq \rho \log S$. □

A.3. Proof of 5.3

Let $M_i \sim \text{Bernoulli}(\delta)$ indicate whether token t_i is modified (i.i.d. across i). For the non-gated case $\rho = 1$, the post-attack indicator at pair $(i, i+1)$ decomposes on (M_i, M_{i+1}):

- $(0,0)$: neither modified; valid with probability 1. Weight $(1-\delta)^2$.
- $(0,1)$: only t_{i+1} modified; by hash uniformity, new $\sigma_k(t_{i+1})$ uniform on $\mathcal{S}$, valid with probability $1/S$. Weight $(1-\delta)\delta$.
- $(1,0)$: symmetric, weight $\delta(1-\delta)$, probability $1/S$.
- $(1,1)$: both modified; still $1/S$ by uniformity. Weight δ^2.

Summing: $\mathbb{E}[\phi^{\text{atk}}] = (1-\delta)^2 + (2\delta(1-\delta) + \delta^2)/S = (1-\delta)^2 + \delta(2-\delta)/S$.

Scaling the deterministic component by gate rate ρ: $\mathbb{E}[\phi^{\text{atk}}] = 1/S + \rho(1-\delta)^2(S-1)/S$.

At the midpoint threshold $\tau = 1/S + \frac{\rho}{2}(S-1)/S$, the attack succeeds when $\rho(1-\delta)^2(S-1)/S < \frac{\rho}{2}(S-1)/S$, i.e. $(1-\delta)^2 < 1/2$. Solving: $\delta^\star = 1 - 1/\sqrt{2}$, independent of S, ρ, n. □

A.4. Proof of 5.4

The state assignment σ_k is a pseudorandom function under the random oracle model. Without k, any polynomial-time distinguisher $\mathcal{A}$ can evaluate σ_k only by querying the oracle; each query reveals $\sigma_k(t)$ for at most one t. Distinguishing watermarked from random sequences requires predicting σ_k on unseen tokens, which by the oracle's uniformity succeeds with probability at most $1/S + \text{negl}$ per query. The adversary's total advantage is bounded by the query budget times $\text{negl}(256)$, i.e. $\text{negl}(256)$.

The gate pattern (g_i) is a function of the LM's entropy profile at the time of generation, which $\mathcal{A}$ can approximate only via its own LM queries; but these do not reveal which positions were gated vs. argmaxed by the generator, since both paths emit tokens with the same marginal distribution conditional on the prefix. Hence (g_i) is not recoverable from tokens alone. □

A.5. Proof of 5.5

Under the null, $X_i \sim \mathrm{Bernoulli}(1/S)$ i.i.d. Under the alternative, $X_i \sim \mathrm{Bernoulli}(\pi_i)$ independently, with $\pi_i = g_i + (1 - g_i)/S$. By the Neyman–Pearson lemma, the uniformly most powerful size-α test for two simple product hypotheses is the log-likelihood ratio test. Computing:

$$\Lambda = \sum_i \log \frac{\pi_i^{X_i}(1-\pi_i)^{1-X_i}}{(1/S)^{X_i}(1-1/S)^{1-X_i}} = \sum_i X_i \log(\pi_i S) + (1 - X_i) \log \tfrac{1-\pi_i}{1-1/S},$$

which matches the theorem statement. □

A.6. Proof of 5.6

Let $\mathcal{S}_k$ denote the set of states derivable from tokens the adversary has oracle access to (its own queries). For any token t' the adversary introduces that is not in $\mathcal{S}_k$, $\sigma_k(t')$ is uniform on $\mathcal{S}$ by the random oracle property. Each modified position j thus contributes an indicator $Z_j \sim \mathrm{Bernoulli}(1/S)$, regardless of the adversary's strategy conditional on no oracle access to $\sigma_k(t'_j)$. Summing over the δn modified positions and accounting for the $(1, 1)$ case as in 5.3 gives the surplus $\Sigma(\delta) = (1/S)\delta(2 - \delta)$. The strategy-independence is a direct consequence of the oracle's uniformity. □

B. Reference Tables

Table 1. Non-gated MCL reference point on the 173-concept benchmark ($\rho = 1$). Llama-3.2-3B-Instruct, greedy decoding, 200-token outputs.

S	WM avg ϕ	Null avg ϕ	TPR	FPR	PPL
5	0.991	0.402	100%	0%	4.98
7	**0.991**	**0.290**	**100%**	**0%**	**4.20**
9	0.984	0.221	100%	0%	5.37
11	0.988	0.197	100%	0%	4.61
15	0.988	0.138	100%	0%	5.11

Table 2. Calibration lookup $S^\star(n, \rho, \alpha)$. Rows are text lengths; columns are (α, ρ) pairs.

	$\alpha = 10^{-3}$		$\alpha = 10^{-6}$	
n	ρ=0.3	ρ=0.5	ρ=0.3	ρ=0.5
100	5	3	13	6
200	3	2	7	3
500	2	2	3	2
1000	2	2	2	2